Hydrogen Fuel Cell Vehicle

Nora Jiang and Matthew Shiley
Professor Carr Everbach

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01

Introduction

What is a HFCV? And why is there one sitting in the hallway?

What you need to know first

Hydrogen (H_2) as a fuel?

- H_2 is an energy carrier (like gasoline)
- Energy is produced through electrolysis
- Stored as a liquid before entering the fuel cell

Why switch to H_2

- H_2 is incredibly energy dense
- Clean in comparison to gas
- Large range before needing to refuel

Inside the Fuel Cell

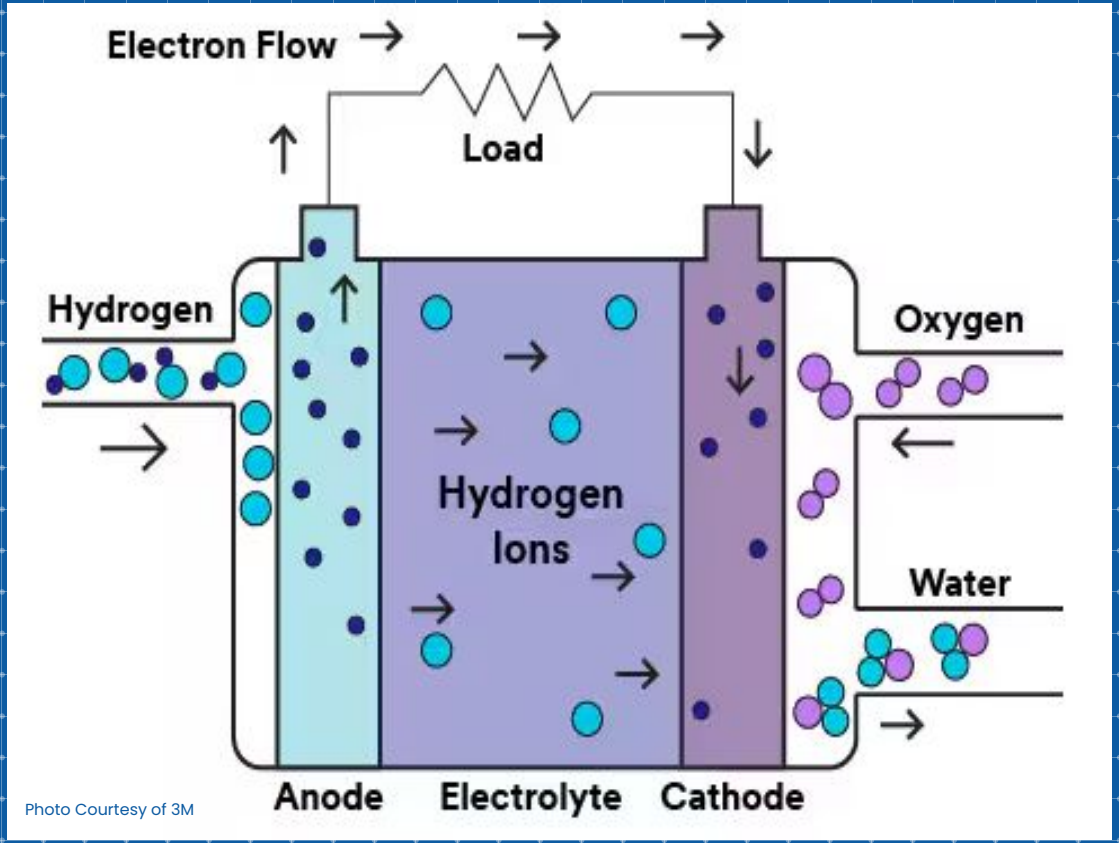


Photo Courtesy of 3M

Our Actual Fuel Cell



Shell Eco Marathon

What Is It?

Competition to make the most energy efficient vehicle

Why Compete?

Gives a target for our work on the vehicle

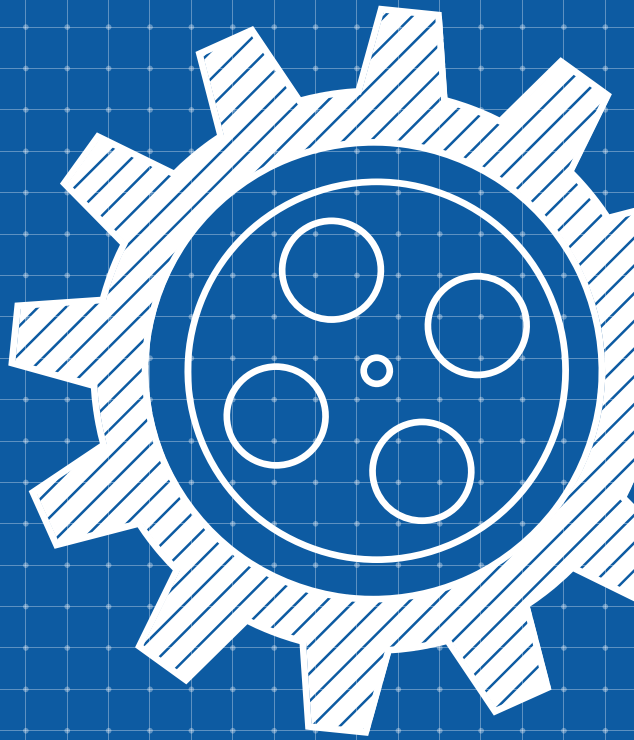


Photo Courtesy of Shell Eco Marathon

02

Current Project

Rebuilding the Chassis, pt. 2



Before This Summer

- Previous vehicle
  - No longer adheres to Shell's safety standards
  - Needs a complete rebuild before competition ready

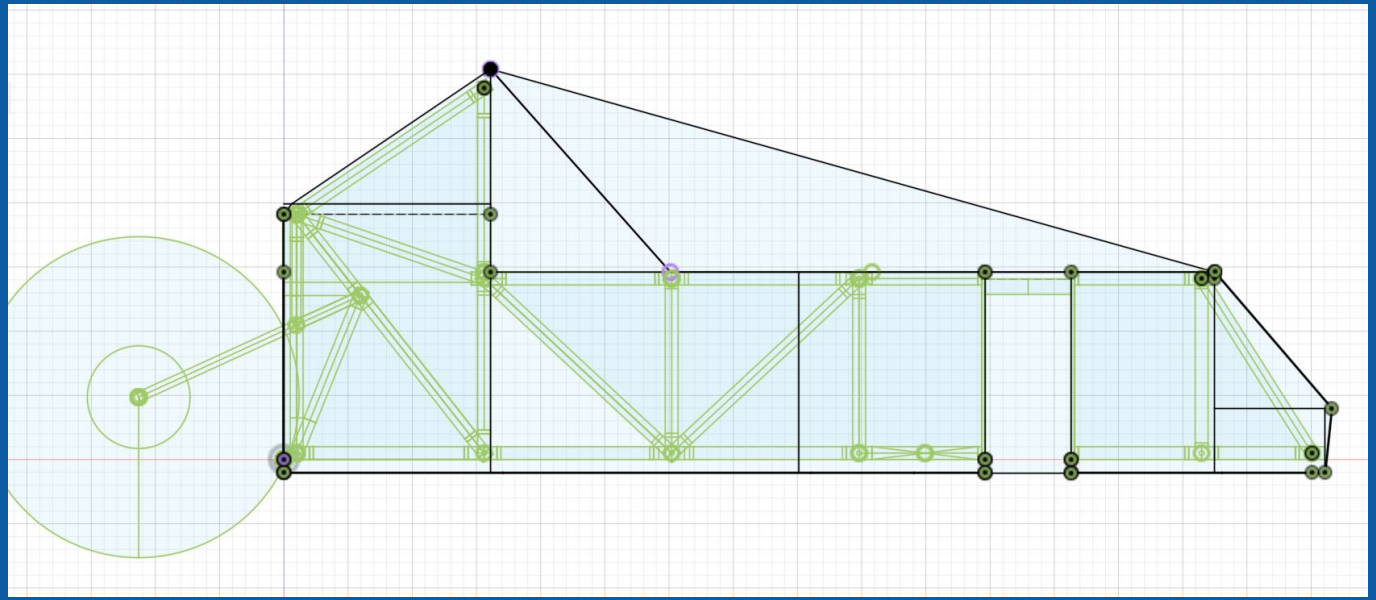
- Revitalization: 2023-2025
  - Materials testing
  - Preliminary Chassis Design
  - Rear Brakes and Suspension system
  - New Braking system
  - Steering components



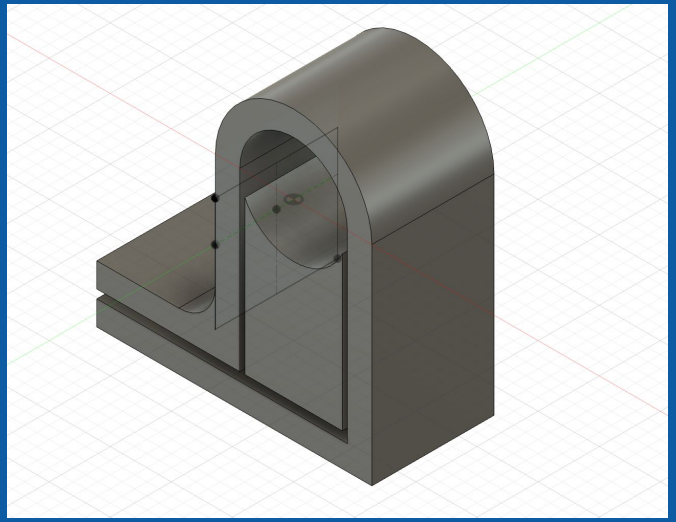
Progress So Far!

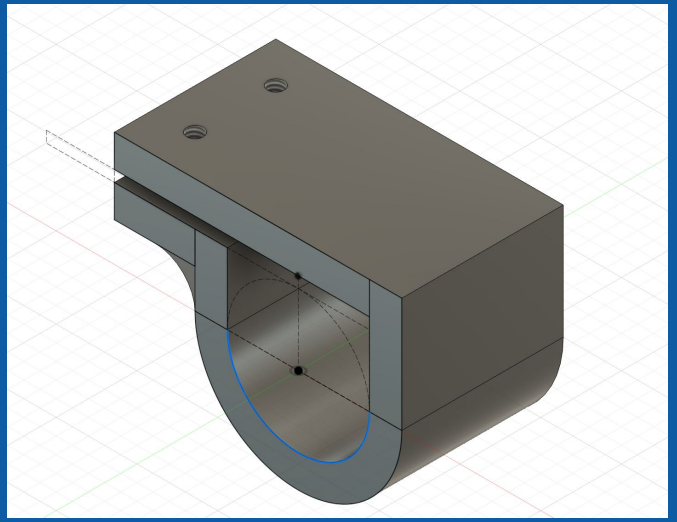


Rough Chassis Design Plan



3D Printed Clips

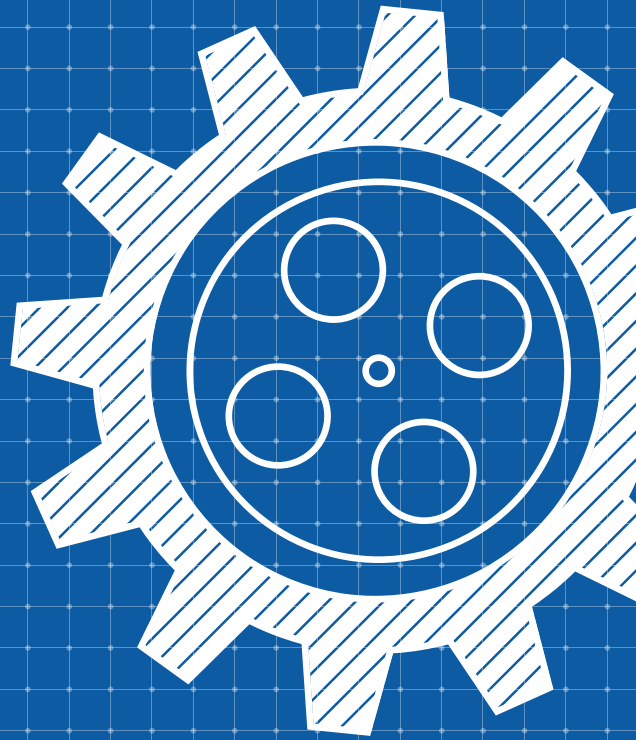




03

Future Considerations

Rebuilding the rest of the HFCV



The Next Year (or so)

Putting it all together

Chassis + Cowl
+
Mechanical
Components

Testing

Testing entire
system to the Shell
Eco Marathon
Requirements

Compete

Transport Team
and Car to Indiana

Thank you for listening

Questions, Comments, Concerns?

njiang1@swarthmore.edu
mshiley1@swarthmore.edu
avasque1@swarthmore.edu
ceverba1@swarthmore.edu